

Development of an Offline-First Point of Sale Mobile Application with Daily Sales Forecasting Using Long Short-Term Memory (LSTM)

Muhammad Kholis¹, Zulfan Khairil Simbolon^{2*}, Azhar³, Juneidy⁴

^{1,2,3} Department of Information and Computer Technology, Politeknik Negeri Lhokseumawe, Jl. Banda Aceh–Medan Km. 280.3 Buketrata, Lhokseumawe 24301, Indonesia

⁴ Study Program of Graphic Multimedia Engineering Technology, Department of Computer and Informatics Engineering, Politeknik Negeri Medan, Jl. Almamater No. 1 USU Campus, Medan 20155, Indonesia

*Corresponding Author: zulfan@pnl.ac.id

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Abstract

Micro, Small, and Medium Enterprises (MSMEs) frequently encounter challenges stemming from internet connectivity dependency in digital cashier systems, as well as high uncertainty in daily sales volume during inventory planning. This research designs and develops an offline-first mobile Point of Sale (POS) application integrating daily sales forecasting features powered by Long Short-Term Memory (LSTM), with an empirical case study at the EatsTEDi canteen, Vocational College, Universitas Gadjah Mada (SV UGM). The client application was engineered using Flutter and the local Isar Database, synchronized bidirectionally with cloud Supabase PostgreSQL, while the forecasting service runs via a containerized REST API (Flask/TensorFlow). Time series modeling was performed on 313 active days of real transaction records using log-return target transformation and benchmarked against an operational suitability threshold ($sMAPE < 20\%$). One-step-ahead forecasting evaluation recorded an average MAE of IDR 667,718.00, RMSE of IDR 897,563.00, and $sMAPE$ of 38.06%, indicating that the model did not reach practical operational acceptance due to data size constraints (212 training sequences) and high volatility in daily transaction patterns. Stability testing across 10 random weight initializations (multi-seed) produced an MAE spread between IDR 495,527.00 and IDR 1,048,967.00 (standard deviation of IDR 145,024.00) and an $sMAPE$ range of 29.16%–44.63% (0 out of 10 seeds achieved the target threshold). Nevertheless, black-box functional testing across 12 core use cases, offline-to-online synchronization verification, and white-box logic path tracing across 7 critical functions achieved a 100% PASS success rate, demonstrating the robust reliability of the offline-first POS architecture in ensuring uninterrupted cashier operations for MSMEs without perpetual internet connectivity.

Keywords: Point of Sale, Daily Sales Forecasting, Long Short-Term Memory, Offline-First, Isar DB, MSME.

1. Introduction

Micro, Small, and Medium Enterprises (MSMEs) represent a vital cornerstone of national economic growth in Indonesia. Nevertheless, the majority of MSME owners still administer their daily operations through manual bookkeeping or disjointed, non-integrated recording workflows [1]. In cashier point-of-sale operations, numerous business owners encounter severe operational bottlenecks caused by continuous dependency on internet connectivity [2]. Pure cloud-based cashier applications frequently fail to function when network connectivity is disrupted, thereby halting payment processing and triggering customer queues. Therefore, an offline-first POS system architecture is urgently required, enabling cashier transactions to be executed locally with instant response times and synchronizing data automatically once network connectivity is restored [3].

Beyond cashier reliability at the checkout counter, another critical operational challenge confronting MSME

managers is the unpredictability of daily sales volumes for supply chain logistics and inventory optimization [4]. The lack of accurate demand estimates exposes businesses to substantial financial loss resulting from stockouts as well as excessive inventory accumulation leading to perishable waste (*overstocking*) [5]. In this regard, time series forecasting provides a strategic computational mechanism for quantitatively projecting operational demands [6].

Prior investigations have extensively evaluated computational intelligence models for retail sales demand forecasting across diverse operational settings. Moraes et al. engineered hybrid stacked and parallel CNN-LSTM models across 500 retail store time series spanning five years, yielding MAPE values of 14.15% and 14.76%, respectively [7]. Mansur et al. proposed a CNN-LSTM architecture incorporating seven exogenous variables (such as weather and public holidays) across six months of retail sales data, achieving a MAPE of 4.16% and outperforming standalone ANN (19.74%) and single LSTM (10.39%) models [8]. Mitra et al. implemented LSTM neural networks with Keras Tuner hyperparameter optimization on commercial retail datasets, evidencing notable error reduction compared to traditional baselines [9]. Yadav and Barde developed a real-time web-based sales prediction pipeline combining LSTM and XGBoost linked to POS registers [10]. In Indonesian retail studies, Sembiring and Sary deployed LSTM for warehouse inventory replenishment, reporting MAPE values of 102–106% compared to 85–88% for Moving Average baselines [11], while Jannah et al. compared LSTM and SARIMA on commercial water refill businesses [12]. Furthermore, Verdiyanto et al. engineered an MSME beverage POS application embedded with Random Forest Regression [13].

Although deep learning architectures such as LSTM are conceptually designed to capture complex non-linear temporal dependencies across sequential data [14], [15], findings by Makridakis et al. in the M5 Competition underscored that simple statistical benchmarks often remain highly competitive against intricate neural networks when applied to short or volatile time series [16]. In micro-enterprise environments characterized by restricted historical records, time series models remain exceptionally sensitive to academic calendar breaks and weight initialization instability. Furthermore, previous studies have rarely benchmarked model robustness against rigorous operational feasibility thresholds or integrated forecasting inference directly into an offline-first mobile POS platform.

To address these empirical and architectural gaps, this study designs and implements an offline-first mobile Point of Sale system built with Flutter and Isar DB connected to Supabase PostgreSQL, integrated with a daily sales forecasting microservice powered by an LSTM neural network deployed via Flask on Hugging Face Spaces. The study conducts an in-depth empirical performance evaluation using both one-step-ahead forecasting and multi-seed stability testing against a practical operational threshold ($sMAPE < 20\%$) using 313 active days of real transaction records from the EatsTEDi SV UGM canteen. System verification was rigorously performed using black-box functional testing, offline synchronization validation, and white-box control path analysis across mission-critical software routines.

2. Research Methods

2.1 Research Workflow

This research was conducted under a quantitative Research and Development (R&D) paradigm. The methodology comprises two interconnected core phases: the design and engineering of the offline-first Point of Sale application, and the development and empirical evaluation of the LSTM time series forecasting model, as depicted in Figure 1.

2.2 Offline-First POS System Architecture

The system architecture was designed to guarantee uninterrupted cashier transactions without requiring persistent network connectivity. The platform consists of four primary components (Figure 1):

- Mobile Client Application: Built with Flutter (Dart programming language) following the Riverpod reactive state management architecture [3], [20]. Local persistence is managed by Isar Database [21], which stores product catalogs, categories, and receipts instantaneously. A background Sync Engine periodically monitors network connectivity; upon detecting an active internet connection, it dispatches unsynchronized local transaction records marked with `is_synced = false` to the cloud database.
- Cloud Infrastructure (Supabase): Utilizes managed Supabase PostgreSQL as a centralized relational database equipped with Row Level Security (RLS) policies [22] and a stored procedure (*Remote Procedure Call*/RPC) named `create_transaction_v4` to execute atomic transaction records and inventory deductions.
- Telemetry and Messaging Services (Firebase): Employs Firebase Cloud Messaging (FCM) for broadcast announcements and low-stock push alerts, and Firebase Analytics for in-app operational telemetry.
- Predictive Inference Service (Hugging Face): The trained LSTM model is served via a Flask REST API on

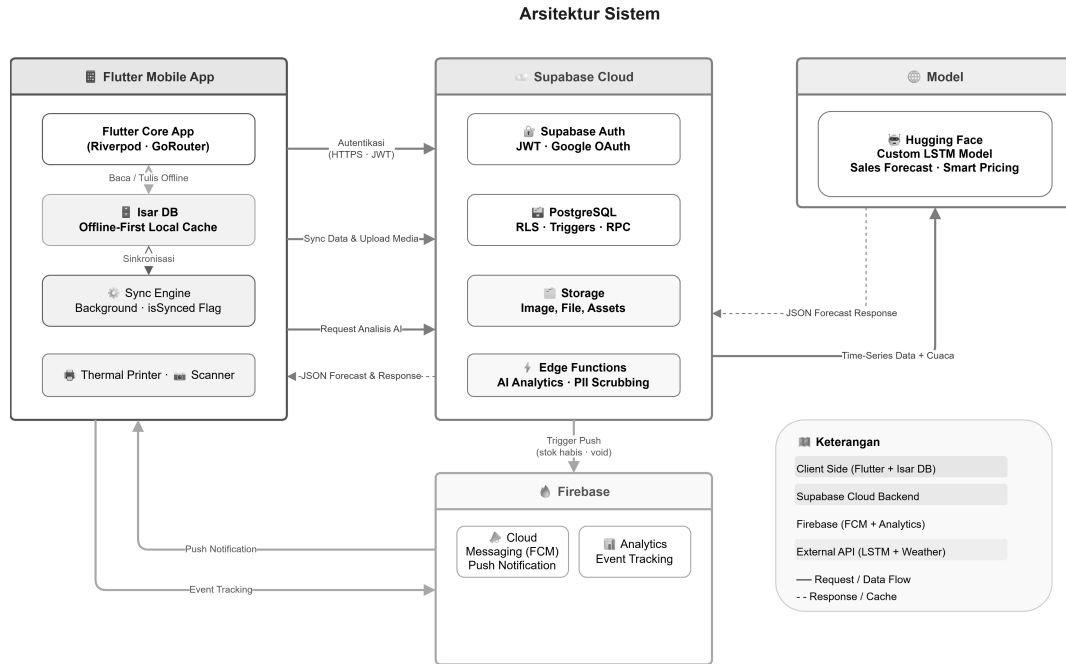


Figure 1. Architecture of the Offline-First Mobile POS System Integrated with LSTM Forecasting Services

Hugging Face Spaces. The mobile client transmits aggregated historical revenue sequences via HTTPS and receives multi-step forecasting projections in structured JSON format.

2.3 Dataset and Data Preprocessing

The empirical dataset comprises authentic transactional records collected from the EatsTEDi SV UGM canteen spanning August 26, 2024, to June 21, 2026, comprising 314 daily records and over 94,000 transaction line items. A representative preview of the daily sales time-series dataset is presented in Table 1.

Table 1. Sample Preview of the Daily Sales Time Series Dataset (EatsTEDi UGM)

Date	Revenue (IDR)	Trx	Qty	DOW	Weekend	Holiday	Ramadan
2024-08-26	1,701,500	113	585	0	0	0	0
2024-08-27	1,746,500	245	611	1	0	0	0
2024-08-28	1,650,500	179	570	2	0	0	0
2024-08-29	1,814,500	219	646	3	0	0	0
2024-08-30	1,233,500	167	453	4	0	0	0
2024-09-02	1,393,500	176	489	0	0	0	0

Across the operational timeline, data preprocessing was systematically conducted through the following stages:

- Active Day Filtering: Retained only operational business days with positive gross revenue (revenue > 0), resulting in 313 active sales days. Academic holiday closures and non-trading weekends were pruned to establish a continuous operational sequence.
- Target Log-Return Transformation: To mitigate non-stationarity and stabilize wide revenue variance, the target series was converted into daily log-returns (r_t):

$$r_t = \ln \left(\frac{Y_t + 1}{Y_{t-1} + 1} \right) \quad (1)$$

where Y_t represents the nominal gross sales revenue on active day t . The +1 offset prevents undefined values during zero-revenue anomalies.

- Cyclical Temporal Features: Extracted day-of-week calendar cycles using sine and cosine trigonometric transformations:

$$\sin_{\text{dow}} = \sin \left(\frac{2\pi \cdot d}{7} \right), \quad \cos_{\text{dow}} = \cos \left(\frac{2\pi \cdot d}{7} \right) \quad (2)$$

where $d \in \{0, 1, \dots, 6\}$ designates the day index within the weekly cycle.

- Normalization and Sequence Construction: Continuous numerical features were scaled into $[0, 1]$ via Min-Max Scaling [23]. Supervised learning samples were generated using a sliding window across 7 active trading days ($look_back = 7$), resulting in an input tensor dimension of 7×8 .
- Chronological Dataset Splitting: Data was partitioned chronologically (time-based split) into 70% training set (212 sequences), 15% validation set (45 sequences), and 15% testing set (46 sequences).

2.4 Long Short-Term Memory (LSTM) Model Architecture

The LSTM recurrent neural network was employed to learn temporal sequential patterns from sales history [14]. Within each LSTM cell at time step t , state transitions are governed by four gating mechanisms:

$$\mathbf{f}_t = \sigma(\mathbf{W}_f \cdot [\mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_f) \quad (3)$$

$$\mathbf{i}_t = \sigma(\mathbf{W}_i \cdot [\mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_i) \quad (4)$$

$$\tilde{\mathbf{C}}_t = \tanh(\mathbf{W}_C \cdot [\mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_C) \quad (5)$$

$$\mathbf{C}_t = \mathbf{f}_t \odot \mathbf{C}_{t-1} + \mathbf{i}_t \odot \tilde{\mathbf{C}}_t \quad (6)$$

$$\mathbf{o}_t = \sigma(\mathbf{W}_o \cdot [\mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_o) \quad (7)$$

$$\mathbf{h}_t = \mathbf{o}_t \odot \tanh(\mathbf{C}_t) \quad (8)$$

where $\mathbf{f}_t, \mathbf{i}_t, \mathbf{o}_t$ denote the forget gate, input gate, and output gate vectors, respectively; $\mathbf{C}_t$ represents the internal cell state; $\mathbf{h}_t$ denotes the hidden state; $\sigma(\cdot)$ is the logistic sigmoid activation; and $\odot$ represents element-wise multiplication (Hadamard product).

The layer configuration of the Hybrid LSTM model is summarized in Algorithm 1, and the training hyperparameters are listed in Table 2. Optimization was carried out using the Adam algorithm [24], complemented by dropout regularization [25] to prevent model overfitting.

Algorithm 1 Construction of Hybrid Residual LSTM Model Using TensorFlow/Keras

Require: Input sequences $\mathbf{X}_t \in \mathbb{R}^{7 \times 8}$ ($look_back = 7, n_features = 8$).

Ensure: Predicted log-return of daily sales revenue $\hat{r}_t \in \mathbb{R}$.

```

1: model ← Sequential()
2: model.add(LSTM(units = 24, input_shape = (7, 8), return_sequences = False))
3: model.add(Dropout(rate = 0.1))
4: model.add(Dense(units = 16, activation = 'relu'))
5: model.add(Dense(units = 1, activation = 'linear'))
6: model.compile(optimizer = Adam(lr = 5 × 10-4), loss = 'mse', metrics = ['mae'])
7: return model

```

Table 2. Training Hyperparameter Specifications for the LSTM Model

Hyperparameter	Value / Specification
Input Tensor Shape	7×8 ($LOOK_BACK = 7$ active days, 8 features)
LSTM Hidden Layer	1 layer, 24 LSTM cell units
Dropout Rate	0.1
Fully Connected Layer	Dense 16 units (ReLU activation)
Output Layer	Dense 1 linear unit (log-return prediction)
Total Trainable Parameters	3,585 parameters
Optimization Algorithm	Adam (learning rate = 5×10^{-4})
Loss Function	Mean Squared Error (MSE)
Batch Size	16
Maximum Epochs	200 (Early Stopping <i>patience</i> =20 & ReduceLROnPlateau)

2.5 Nominal Reconstruction and Multi-Step Projection

The predicted log-return $\hat{r}_t$ was mapped back into nominal currency scale (IDR, $\hat{Y}_t$) via inverse exponential reconstruction:

$$\hat{Y}_t = \exp(\ln(Y_{t-1} + 1) + \hat{r}_t) - 1 \quad (9)$$

For recursive multi-step forecasting extending up to 7 active business days ahead, three safety guards were enforced:

- Clipping: Bound forecasted log-returns within the 10th and 90th empirical percentiles of the training distribution to prevent extreme deviations.
- Damping: Scaled projected variations by an exponential decay factor ϕ^{h-1} ($\phi = 0.7$) as the step horizon h increases.
- Safety Rail: Clamped projected nominal sales revenue to never exceed the historical upper bound (revenue ceiling).

2.6 Evaluation Metrics and Feasibility Benchmark

Forecasting accuracy was examined under a one-step-ahead protocol utilizing three complementary evaluation metrics [6], [26]:

$$\text{MAE} = \frac{1}{N} \sum_{t=1}^N |Y_t - \hat{Y}_t| \quad (10)$$

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{t=1}^N (Y_t - \hat{Y}_t)^2} \quad (11)$$

$$\text{sMAPE} = \frac{100\%}{N} \sum_{t=1}^N \frac{|Y_t - \hat{Y}_t|}{(|Y_t| + |\hat{Y}_t|)/2} \quad (12)$$

where Y_t denotes actual gross revenue, $\hat{Y}_t$ is the forecasted revenue, and N represents the test sample size ($N = 46$). Symmetric MAPE (sMAPE) serves as the primary benchmark because it provides bounded, scale-free percentage errors without diverging when encountering small sales values. The practical operational threshold was set at $\text{sMAPE} < 20\%$.

3. Results and Discussion

3.1 Evaluation of LSTM Forecasting Model Against Practical Threshold

Model performance on the test dataset (46 active days) was assessed across two rigorous evaluation scenarios. The first scenario benchmarked the one-step-ahead forecasting accuracy of the Hybrid LSTM model. The empirical evaluation metrics are summarized in Table 3.

Table 3. Hybrid LSTM Model Performance on Test Dataset (One-Step-Ahead)

Model / Architecture	MAE (IDR)	RMSE (IDR)	sMAPE (%)
Hybrid LSTM (10-Seed Average)	667,718	897,563	38.06%
Practical Feasibility Threshold	—	—	< 20.00%

As detailed in Table 3, the Hybrid LSTM model yielded an average MAE of IDR 667,718.00, RMSE of IDR 897,563.00, and sMAPE of 38.06%. The sMAPE metric substantially exceeded the pre-established 20% operational feasibility benchmark. The visual comparison between actual revenue and model forecasts is illustrated in Figure 2.

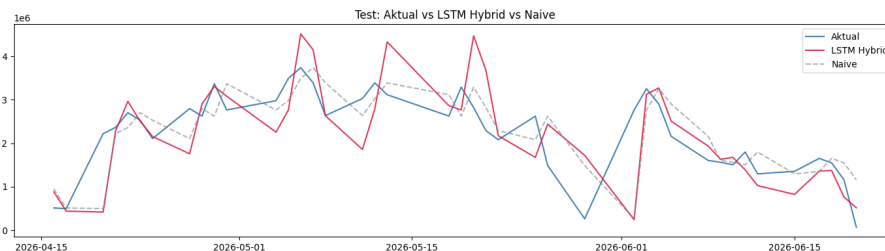


Figure 2. Time Series Comparison of Actual Canteen Revenue Against Hybrid LSTM Model Forecasts

The trajectory in Figure 2 highlights that the LSTM model exhibited pronounced over-smoothing. While the model successfully approximated medium-term central revenue tendencies, it was unable to capture sudden, high-magnitude sales peaks induced by campus events and irregular academic schedules.

3.2 Model Stability Testing via Multi-Seed Evaluation

The second evaluation scenario tested model stability against random weight initialization variations. The training pipeline was executed 10 times under identical parameters using distinct random seeds ($seed \in \{0, 1, \dots, 9\}$). Descriptive performance statistics across the 10 runs are provided in Table 4.

Table 4. Hybrid LSTM Performance Across Multi-Seed Stability Testing (10 Runs)

Evaluation Metric	Mean	Std. Deviation ($\pm$)	Range (Min – Max)
MAE (IDR)	667,718	145,024	495,527 – 1,048,967
RMSE (IDR)	897,563	165,804	703,721 – 1,320,587
sMAPE (%)	38.06%	3.96%	29.16% – 44.63%

The empirical results in Table 4 reveal considerable variance across seeds, exhibiting a standard deviation of IDR 145,024.00 in MAE and 3.96% in sMAPE. The worst-performing initialization (MAE IDR 1,048,967.00) recorded an error more than double that of the best initialization (MAE IDR 495,527.00). Crucially, none of the 10 seeds reached the target sMAPE threshold of $< 20\%$ (the top-performing run achieved 29.16% on Seed 7). The distribution of MAE values across runs is depicted as a boxplot in Figure 3.

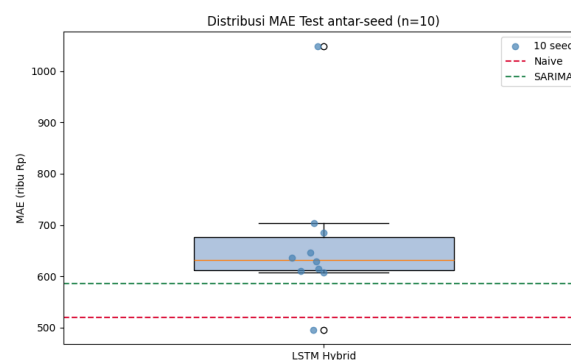


Figure 3. MAE Error Distribution of Hybrid LSTM Across Multi-Seed Stability Runs (10 Iterations)

This dispersion underscores that training neural networks on modest sequential datasets (212 training samples) is inherently prone to initialization instability (*seed variance*). Consequently, reporting single-run performance for deep forecasting architectures on small datasets introduces significant reporting bias. A concise synthesis of both evaluation scenarios is presented in Table 5.

Table 5. Summary of Empirical Evaluation Scenarios for the Forecasting Model

Scenario	Evaluation Objective	Threshold Met?	Key Empirical Finding
1	Measure one-step-ahead accuracy against benchmark (sMAPE $< 20\%$)	No	Mean MAE IDR 667,718.00 and sMAPE 38.06%, exceeding operational threshold.
2	Assess training stability across 10 randomized seeds	No	Wide MAE spread (IDR 495,527–1,048,967) with standard deviation IDR 145,024; 0 of 10 seeds satisfied sMAPE $< 20\%$.

3.3 Mobile Point of Sale Interface Implementation

The mobile POS application was deployed to Android mobile hardware featuring a modern, intuitive user interface. The system delivers complete cashier transaction workflows, catalog management, inventory adjustments, daily revenue dashboards, and a specialized Smart Analytics screen presenting 7-day predictive projections generated by the LSTM inference microservice (Figure 4).

The Smart Analytics module displays projected gross sales, multi-horizon trend charts with ceiling limits (safety rails), and cold-start detection banners that advise store operators when sales history has not yet reached the required 30-day warm-up baseline.

3.4 Functional System Testing (Black Box Testing)

Comprehensive black-box verification was executed across 12 core operational use cases covering both Owner and Cashier roles in compliance with software testing standards [17], [18], [19]. Every test case completed successfully with a 100% PASS outcome (Table 6).

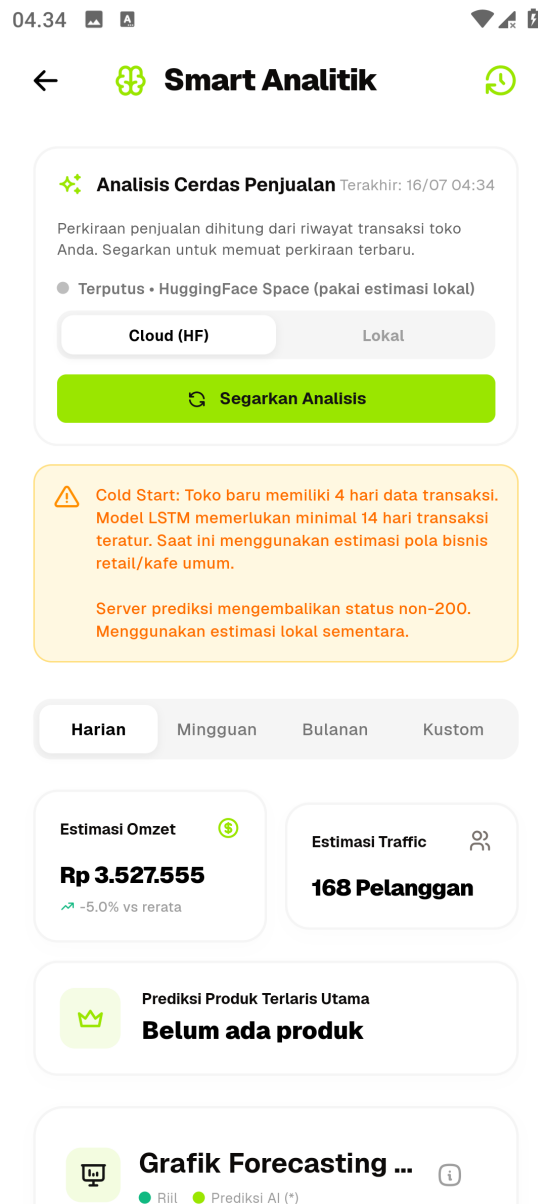


Figure 4. User Interface of the Smart Analytics Module in the Mobile POS Application

Table 6. Black Box Test Outcomes Across Core Mobile POS Features

No	Feature / Test Case	Testing Action	Expected Result	Status
1	Merchant Registration	Enter merchant profile & store credentials	Account registered on cloud PostgreSQL	PASS
2	User Authentication	Submit valid credentials	Dashboard access matched to user role	PASS
3	Checkout Transaction	Select items & confirm payment	Local commit & immediate stock deduction	PASS
4	Product Management	Create, update, and delete menu items	Catalog updated instantly in local DB	PASS
5	Category Organization	Organize menu category taxonomies	Categories saved and displayed at register	PASS
6	Inventory Stock Audit	Review stock adjustment history log	Accurate records of additions/deductions	PASS
7	Sales Dashboard	Review daily turnover summaries	Valid aggregation & sales charts	PASS
8	Smart Analytics	Request 7-day sales projections	Projections displayed with bounds	PASS
9	Staff Notifications	Dispatch broadcast message to staff	FCM notification received on cashier app	PASS
10	Data Synchronization	Restore internet connectivity	Queued transactions synced to cloud	PASS
11	Staff Management	Configure cashier accounts & permissions	Cashier role verified and updated	PASS
12	Store Profile	Update business name, address, and logo	Profile propagated across all modules	PASS

3.5 Evaluation of Offline-First Synchronization Mechanics

In-depth testing of the offline-first architecture was conducted to ensure data integrity during offline-to-online transitions. Results are presented in Table 7.

Table 7. Validation Results of Offline-First Synchronization (Isar DB to Supabase)

No	Test Scenario	Observed System Behavior	Status
1	Checkout during offline state	Receipt saved in local Isar DB with <code>is_synced = false</code> ; inventory reduced locally.	PASS
2	View sales history offline	Instant retrieval of local sales without network latency or UI blocking.	PASS
3	Reconnection to internet	Sync Engine detects online status and uploads queued receipts via Supabase RPC.	PASS
4	Verify cloud data consistency	Transactions committed to Supabase without duplicates; local flag updated to <code>is_synced = true</code> .	PASS

These findings demonstrate that embedding Isar DB provides near-instant cashier responsiveness while preserving total transactional integrity during network outages.

3.6 Code Logic Path Testing (White Box Testing)

White-box unit testing was conducted to verify critical conditional branches in the Python forecasting API service (Table 8).

Table 8. White Box Test Verification of Critical Forecasting API Functions

No	Function / Target Path	Input Test Condition	Expected Execution Outcome	Status
1	<code>by_dow</code> (Seasonal-Naive)	Historical day-of-week data $\geq k$	Computes mean across recent k matching days	PASS
2	<code>by_dow</code> (Partial Branch)	Historical day-of-week data $< k$	Computes mean across all available matching days	PASS
3	<code>by_dow</code> (Fallback Path)	Day-of-week completely missing	Falls back to mean of last 10 active days	PASS
4	<code>smape</code> (Zero Denominator)	Actual and forecast values are zero	Denominator defaulted to 1; zero-division avoided	PASS
5	<code>forecast</code> (Clipping Guard)	Log-return exceeds percentile bound	Value clipped at empirical upper percentile	PASS
6	<code>forecast</code> (Damping Decay)	Forecast step $h > 1$	Variance decayed by multiplicative factor ϕ^{h-1}	PASS
7	<code>forecast</code> (Safety Rail)	Forecast exceeds historical max	Nominal forecast clamped to historical ceiling	PASS

The white-box verification confirmed that all edge case exception routines and safety clamping logic functioned without runtime errors.

3.7 Discussion and Findings Positioning

Empirical findings demonstrate that the forecasting shortfall ($\text{sMAPE} = 38.06\%$) stems directly from the limited volume of historical training records (313 active days, 212 training sequences) paired with the sharp demand volatility characteristic of campus cafeteria environments. This aligns with findings from Makridakis et al. during the M5 Competition, where deep neural networks provided minimal competitive advantage over simpler statistical heuristics on brief time series lacking multi-year cyclical repetitions [16]. Furthermore, eliminating non-trading academic holidays (revenue = 0) created irregular calendar gaps between active business days, challenging the continuous time-step assumptions of recurrent neural architectures [5].

Nonetheless, this research contributes key engineering and methodological insights to applied predictive software development:

- Multi-Seed Evaluation Transparency: Discloses neural network initialization variance through rigorous reporting of standard deviations and error bounds.
- Validity of sMAPE Metric: Prevents exploding percentage error artifacts encountered by standard MAPE when analyzing small-enterprise sales with near-zero baselines.
- Resilient Offline-First Architecture: Demonstrates the practical feasibility of pairing local Isar DB persistence with Supabase PostgreSQL to achieve zero cashier downtime regardless of connectivity status.

4. Conclusion and Future Work

4.1 Conclusion

This research successfully engineered an offline-first mobile Point of Sale application utilizing Flutter, Isar Database, Supabase PostgreSQL, and a containerized LSTM forecasting REST microservice hosted on Hugging Face Spaces. Black-box functional validation across 12 primary use cases, offline synchronization verification, and white-box unit tests all achieved a 100% PASS rate, confirming the system's ability to ensure zero downtime and near-instant checkout operations for MSMEs without continuous network connectivity. Regarding predictive modeling, empirical evaluation of the Hybrid LSTM model on 313 active days yielded an average MAE of IDR 667,718.00, RMSE of IDR 897,563.00, and sMAPE of 38.06%, exceeding the practical feasibility threshold ($sMAPE < 20\%$). Multi-seed stability testing demonstrated substantial variance across runs (MAE IDR 495,527.00–1,048,967.00; sMAPE 29.16%–44.63%), indicating that deep recurrent architectures remain highly sensitive to weight initialization and require larger datasets to achieve stable convergence in micro-retail settings.

4.2 Future Work

Future research should expand data collection horizons across three to five operational years to enable deep sequence models to capture multi-year seasonal cycles effectively. Incorporating elapsed calendar gap features between non-consecutive trading days could alleviate temporal distortions caused by academic breaks. Additionally, future initiatives may explore transfer learning or multi-tenant global architectures across comparable retail categories to mitigate cold-start limitations, as well as investigate hybrid combinations (e.g., SARIMA-LSTM) that unify linear seasonal modeling with non-linear adaptation. Finally, enabling cryptographic encryption on the local Isar Database is recommended to further strengthen device-level data protection.

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